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DES SCIENCES

Faculté des Sciences et Technologies (FST)

Rapport du travail de Laboratoire N° 5_Administration Système

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Niveau : L3

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L'objectif de ce TD est de :

1. sécuriser un serveur Linux
2. protéger les accès SSH
3. filtrer le trafic réseau
4. bloquer les attaques automatisées
5. analyser la sécurité d'un système
6. installer un outil d'audit de sécurité
7. interpréter un rapport d'audit
8. identifier les failles potentielles
9. proposer des améliorations de sécurité

Analyse initiale du serveur

- Vérifier les ports ouverts

```
donsam@donsam:~$ ss -tln
```

Netid	State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port	Process
udp	UNCONN	0	0	127.0.0.54:53	0.0.0.0:*	
udp	UNCONN	0	0	127.0.0.53:10:53	0.0.0.0:*	
udp	UNCONN	0	0	192.168.2.141:53	0.0.0.0:*	
udp	UNCONN	0	0	[fe80::a00:27ff:fe89:f45f]:546	:::*	
tcp	LISTEN	0	4096	0.0.0.0:2222	0.0.0.0:*	
tcp	LISTEN	0	4096	127.0.0.54:53	0.0.0.0:*	
tcp	LISTEN	0	4096	127.0.0.53:10:53	0.0.0.0:*	
tcp	LISTEN	0	4096	:::2222	:::*	

```
donsam@donsam:~$
```

- Vérifier les services actifs

apparmor.service	loaded active exited	Load AppArmor profiles
apport.service	loaded active exited	automatic crash report generation
blk-availability.service	loaded active exited	Availability of block devices
console-setup.service	loaded active exited	Set console font and keymap
cron.service	loaded active running	Regular background program processing daemon
dbus.service	loaded active running	D-Bus System Message Bus
fail2ban.service	loaded active running	Fail2Ban Service
finalrd.service	loaded active exited	Create final runtime dir for shutdown pivot root
getty@tty1.service	loaded active running	Getty on tty1
keyboard-setup.service	loaded active exited	Set the console keyboard layout
kmod-static-nodes.service	loaded active exited	Create List of Static Device Nodes
lvm2-monitor.service	loaded active exited	Monitoring of LVM2 mirrors, snapshots etc. using
ModemManager.service	loaded active running	Modem Manager
multipathd.service	loaded active running	Device-Mapper Multipath Device Controller
nftables.service	loaded active exited	nftables
plymouth-quit-wait.service	loaded active exited	Hold until boot process finishes up
plymouth-quit.service	loaded active exited	Terminate Plymouth Boot Screen
plymouth-read-write.service	loaded active exited	Tell Plymouth To Write Out Runtime Data
polkit.service	loaded active running	Authorization Manager
rsyslog.service	loaded active running	System Logging Service
setvtrgb.service	loaded active exited	Set console scheme
snappy.service	loaded active exited	Load AppArmor profiles managed internally by
snappy-seeded.service	loaded active exited	Wait until snappy is fully seeded
ssh.service	loaded active running	OpenBSD Secure Shell server
sysstat.service	loaded active exited	Resets System Activity Logs
systemd-binfmt.service	loaded active exited	Set Up Additional Binary Formats
systemd-fsck@dev-disk-by\x2duuid-1699624f\x2dbff\x2d4993\x2da14b\x2d27082403679c.service	loaded active exited	File System Check on /dev/disk/by-uuid/1699624f
systemd-journal-flush.service	loaded active exited	Flush Journal to Persistent Storage
systemd-journald.service	loaded active running	Journal Service
systemd-logind.service	loaded active running	User Login Management
systemd-modules-load.service	loaded active exited	Load Kernel Modules
systemd-networkd-wait-online.service	loaded active exited	Wait for Network to be Configured
systemd-networkd.service	loaded active running	Network Configuration
systemd-random-seed.service	loaded active exited	Load/Save OS Random Seed
systemd-remount-fs.service	loaded active exited	Remount Root and Kernel File Systems
systemd-resolved.service	loaded active running	Network Name Resolution
systemd-sysctl.service	loaded active exited	Apply Kernel Variables
systemd-timesyncd.service	loaded active running	Network Time Synchronization
systemd-tmpfiles-setup-dev-early.service	loaded active exited	Create Static Device Nodes in /dev gracefully
systemd-tmpfiles-setup-dev.service	loaded active exited	Create Static Device Nodes in /dev
systemd-tmpfiles-setup.service	loaded active exited	Create Volatile Files and Directories
systemd-udev-trigger.service	loaded active exited	Coldplug All udev Devices
systemd-udevd.service	loaded active running	Rule-based Manager for Device Events and Files
systemd-update-utmp.service	loaded active exited	Record System Boot/Shutdown in UTMP
systemd-user-sessions.service	loaded active exited	Permit User Sessions
udisks2.service	loaded active running	Disk Manager
ufw.service	loaded active exited	Uncomplicated firewall
unattended-upgrades.service	loaded active running	Unattended Upgrades Shutdown

- Vérifier les utilisateurs

```
donsam@donsam:~$ cat /etc/passwd
```

root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailng List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:/:nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd/bin/false
messagebus:x:101:102:/:nonexistent:/usr/sbin/nologin
systemd-resolve:x:992:992:systemd Resolver:/:/usr/sbin/nologin
pollinate:x:102:1:/:var/cache/pollinate/bin/false
polkitd:x:991:991:User for polkitd:/:/usr/sbin/nologin
syslog:x:103:104:/:nonexistent:/usr/sbin/nologin
uuidd:x:104:105:/:run/uuidd:/usr/sbin/nologin
tcpdump:x:105:107:/:nonexistent:/usr/sbin/nologin
tss:x:106:108:TPM software stack,,,:/var/lib/tpm/bin/false
landscape:x:107:109:/:var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
usbmux:x:108:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
donsam:x:1000:1000:Donsam:/home/donsam:/bin/bash
sshd:x:109:65534:/:run/sshd:/usr/sbin/nologin

Sécurisation de SSH

```
GNU nano 7.2 /etc/ssh/sshd_config

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

# When systemd socket activation is used (the default), the socket
# configuration must be re-generated after changing Port, AddressFamily, or
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
Port 2222
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
MaxAuthTries 3
#MaxSessions 10

#PubkeyAuthentication yes
PermitRootLogin no
PasswordAuthentication no
AllowUsers donsam
# Expect .ssh/authorized_keys2 to be disregarded by default in future.
```

```

donsam@donsam:~$ sudo systemctl restart ssh
donsam@donsam:~$ 1234
1234: command not found
donsam@donsam:~$ ssh -p 2222 donsam@192.168.2.141
The authenticity of host '[192.168.2.141]:2222 ([192.168.2.141]:2222)' can't be established.
ED25519 key fingerprint is SHA256:UXb5Y75wN1EYxALu17enAC7TtPpnrj0xdX/2I1l6EY.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[192.168.2.141]:2222' (ED25519) to the list of known hosts.
donsam@192.168.2.141's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-107-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of sam. 04 avril 2026 15:31:41 UTC

System load:            0.0
Usage of /:             29.3% of 23.45GB
Memory usage:           6%
Swap usage:             0%
Processes:              121
Users logged in:        1
IPv4 address for enp0s3: 192.168.2.141
IPv6 address for enp0s3: fdbb:16fa:e40f:30:a00:27ff:fe89:f45f
IPv6 address for enp0s3: 2803:9810:a0e1:9530:a00:27ff:fe89:f45f

La maintenance de sécurité étendue pour Applications n'est pas activée.

0 mise à jour peut être appliquée immédiatement.

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez https://ubuntu.com/esm ou exécutez : sudo pro status

Last login: Sun Mar 29 21:14:21 2026 from 192.168.1.123
donsam@donsam:~$

```

Configuration du Firewall

- Installation UFW

```

Last login: Sun Mar 29 21:14:21 2026 from 192.168.1.123
donsam@donsam:~$ sudo apt install ufw -y
[sudo] password for donsam:
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
ufw est déjà la version la plus récente (0.36.2-6).
0 mis à jour, 0 nouvellement installés, 0 à enlever et 0 non mis à jour.
donsam@donsam:~$ sudo ufw allow 2222/tcp

```

- Autorisation des ports nécessaires

```

0 mis à jour, 0 nouvellement installés, 0 à enlever et 0 non mis à jour.
donsam@donsam:~$ sudo ufw allow 2222/tcp
Rule added
Rule added (v6)
donsam@donsam:~$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
donsam@donsam:~$ sudo ufw allow 433/tcp
Rule added
Rule added (v6)
donsam@donsam:~$

```

- Activation de firewall et vérification des règles actives

```
donsam@donsam:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
donsam@donsam:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: allow (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22 ALLOW IN Anywhere
80 ALLOW IN Anywhere
443 ALLOW IN Anywhere
22/tcp ALLOW IN Anywhere
2222/tcp ALLOW IN Anywhere
80/tcp ALLOW IN Anywhere
443/tcp ALLOW IN Anywhere
22 (v6) ALLOW IN Anywhere (v6)
80 (v6) ALLOW IN Anywhere (v6)
443 (v6) ALLOW IN Anywhere (v6)
22/tcp (v6) ALLOW IN Anywhere (v6)
2222/tcp (v6) ALLOW IN Anywhere (v6)
80/tcp (v6) ALLOW IN Anywhere (v6)
443/tcp (v6) ALLOW IN Anywhere (v6)
```

Protection contre brute force (Fail2ban)

```
donsam@donsam:~$ sudo apt install fail2ban -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
fail2ban est déjà la version la plus récente (1.0.2-3ubuntu0.1).
0 mis à jour, 0 nouvellement installés, 0 à enlever et 0 non mis à jour.
donsam@donsam:~$
```

```
donsam@donsam:~$ sudo systemctl restart fail2ban
donsam@donsam:~$ sudo fail2ban-client status
Status
|- Number of jail:      1
| - Jail list:  sshd
donsam@donsam:~$
```

Analyse des logs

```
2026-04-04T15:30:16.447251+00:00 donsam sudo: pam_unix(sudo:session): session closed for user root
2026-04-04T15:30:53.264790+00:00 donsam sudo: donsam : TTY=tty1 ; PWD=/home/donsam ; USER=root ; COMMAND=/usr/bin/sy
2026-04-04T15:30:53.267272+00:00 donsam sudo: pam_unix(sudo:session): session opened for user root(uid=0) by donsam(uid
2026-04-04T15:30:53.286521+00:00 donsam sshd[1121]: Received signal 15; terminating.
2026-04-04T15:30:53.374932+00:00 donsam sshd[1135]: Server listening on 0.0.0.0 port 2222.
2026-04-04T15:30:53.375469+00:00 donsam sshd[1135]: Server listening on :: port 2222.
2026-04-04T15:30:53.381332+00:00 donsam sudo: pam_unix(sudo:session): session closed for user root
2026-04-04T15:31:41.308686+00:00 donsam sshd[1145]: Accepted password for donsam from 192.168.2.141 port 60846 ssh2
2026-04-04T15:31:41.314027+00:00 donsam sshd[1145]: pam_unix(sshd:session): session opened for user donsam(uid=1000) by
2026-04-04T15:31:41.330185+00:00 donsam systemd-logind[768]: New session 7 of user donsam.
2026-04-04T15:35:01.808915+00:00 donsam CRON[1213]: pam_unix(cron:session): session opened for user root(uid=0) by root
2026-04-04T15:35:01.836425+00:00 donsam CRON[1213]: pam_unix(cron:session): session closed for user root
2026-04-04T15:38:59.502488+00:00 donsam sudo: donsam : TTY=pts/0 ; PWD=/home/donsam ; USER=root ; COMMAND=/usr/bin/ap
2026-04-04T15:38:59.528390+00:00 donsam sudo: pam_unix(sudo:session): session opened for user root(uid=0) by donsam(uid
2026-04-04T15:39:02.551289+00:00 donsam sudo: pam_unix(sudo:session): session closed for user root
2026-04-04T15:40:52.421746+00:00 donsam sudo: donsam : TTY=pts/0 ; PWD=/home/donsam ; USER=root ; COMMAND=/usr/sbin/u
2026-04-04T15:40:52.448706+00:00 donsam sudo: pam_unix(sudo:session): session opened for user root(uid=0) by donsam(uid
2026-04-04T15:40:52.892376+00:00 donsam sudo: pam_unix(sudo:session): session closed for user root
2026-04-04T15:41:15.626480+00:00 donsam sudo: donsam : TTY=pts/0 ; PWD=/home/donsam ; USER=root ; COMMAND=/usr/sbin/u
2026-04-04T15:41:15.631772+00:00 donsam sudo: pam_unix(sudo:session): session opened for user root(uid=0) by donsam(uid
2026-04-04T15:41:16.040448+00:00 donsam sudo: pam_unix(sudo:session): session closed for user root
2026-04-04T15:41:34.397564+00:00 donsam sudo: donsam : TTY=pts/0 ; PWD=/home/donsam ; USER=root ; COMMAND=/usr/sbin/u
2026-04-04T15:41:34.398255+00:00 donsam sudo: pam_unix(sudo:session): session opened for user root(uid=0) by donsam(uid
2026-04-04T15:41:34.793158+00:00 donsam sudo: pam_unix(sudo:session): session closed for user root
2026-04-04T15:43:20.795191+00:00 donsam sudo: donsam : TTY=pts/0 ; PWD=/home/donsam ; USER=root ; COMMAND=/usr/sbin/u
2026-04-04T15:43:20.798760+00:00 donsam sudo: pam_unix(sudo:session): session opened for user root(uid=0) by donsam(uid
2026-04-04T15:43:26.977823+00:00 donsam sudo: pam_unix(sudo:session): session closed for user root
2026-04-04T15:44:06.573676+00:00 donsam sudo: donsam : TTY=pts/0 ; PWD=/home/donsam ; USER=root ; COMMAND=/usr/sbin/u
2026-04-04T15:44:06.578872+00:00 donsam sudo: pam_unix(sudo:session): session opened for user root(uid=0) by donsam(uid
2026-04-04T15:44:06.806758+00:00 donsam sudo: pam_unix(sudo:session): session closed for user root
2026-04-04T15:45:01.862288+00:00 donsam CRON[1453]: pam_unix(cron:session): session opened for user root(uid=0) by root
2026-04-04T15:45:01.898210+00:00 donsam CRON[1453]: pam_unix(cron:session): session closed for user root
2026-04-04T15:45:23.216542+00:00 donsam sudo: donsam : TTY=pts/0 ; PWD=/home/donsam ; USER=root ; COMMAND=/usr/bin/ap
2026-04-04T15:45:23.219631+00:00 donsam sudo: pam_unix(sudo:session): session opened for user root(uid=0) by donsam(uid
2026-04-04T15:45:25.229753+00:00 donsam sudo: pam_unix(sudo:session): session closed for user root
2026-04-04T15:51:02.877609+00:00 donsam sudo: donsam : TTY=pts/0 ; PWD=/home/donsam ; USER=root ; COMMAND=/usr/bin/ca
2026-04-04T15:51:02.909207+00:00 donsam sudo: pam_unix(sudo:session): session opened for user root(uid=0) by donsam(uid
2026-04-04T15:51:02.909503+00:00 donsam sudo: pam_unix(sudo:session): session closed for user root
2026-04-04T15:54:29.676226+00:00 donsam sudo: donsam : TTY=pts/0 ; PWD=/home/donsam ; USER=root ; COMMAND=/usr/bin/na
2026-04-04T15:54:29.678879+00:00 donsam sudo: pam_unix(sudo:session): session opened for user root(uid=0) by donsam(uid
2026-04-04T15:55:01.923398+00:00 donsam CRON[1523]: pam_unix(cron:session): session opened for user root(uid=0) by root
2026-04-04T15:55:01.939025+00:00 donsam CRON[1523]: pam_unix(cron:session): session closed for user root
2026-04-04T15:56:14.251956+00:00 donsam sudo: pam_unix(sudo:session): session closed for user root
2026-04-04T15:56:45.341419+00:00 donsam sudo: donsam : TTY=pts/0 ; PWD=/home/donsam ; USER=root ; COMMAND=/usr/bin/sy
2026-04-04T15:56:45.343717+00:00 donsam sudo: pam_unix(sudo:session): session opened for user root(uid=0) by donsam(uid
2026-04-04T15:56:45.841391+00:00 donsam sudo: pam_unix(sudo:session): session closed for user root
2026-04-04T15:57:13.069467+00:00 donsam sudo: donsam : TTY=pts/0 ; PWD=/home/donsam ; USER=root ; COMMAND=/usr/bin/fa
2026-04-04T15:57:13.072398+00:00 donsam sudo: pam_unix(sudo:session): session opened for user root(uid=0) by donsam(uid
2026-04-04T15:57:13.268708+00:00 donsam sudo: pam_unix(sudo:session): session closed for user root
```

```
donsam@donsam:~$ grep "Failed password" /var/log/auth.log
donsam@donsam:~$ cat /var/log/fail2ban.log
```

```

2026-04-04 14:11:30,264 fail2ban.jail [774]: INFO Creating new jail 'sshd'
2026-04-04 14:11:30,926 fail2ban.jail [774]: INFO Jail 'sshd' uses systemd {}
2026-04-04 14:11:30,937 fail2ban.jail [774]: INFO Initiated 'systemd' backend
2026-04-04 14:11:30,939 fail2ban.filter [774]: INFO maxLines: 1
2026-04-04 14:11:30,957 fail2ban.filtersystemd [774]: INFO [sshd] Added journal match for: '_SYSTEMD_UNIT=sshd.servi
2026-04-04 14:11:30,957 fail2ban.filter [774]: INFO maxRetry: 5
2026-04-04 14:11:30,957 fail2ban.filter [774]: INFO findtime: 600
2026-04-04 14:11:30,957 fail2ban.actions [774]: INFO banTime: 600
2026-04-04 14:11:30,957 fail2ban.filter [774]: INFO encoding: UTF-8
2026-04-04 14:11:30,964 fail2ban.jail [774]: INFO Jail 'sshd' started
2026-04-04 14:11:30,989 fail2ban.filtersystemd [774]: INFO [sshd] Jail is in operation now (process new journal entr
2026-04-04 15:04:19,065 fail2ban.server [830]: INFO -----
2026-04-04 15:04:19,067 fail2ban.server [830]: INFO Starting Fail2ban v1.0.2
2026-04-04 15:04:19,069 fail2ban.observer [830]: INFO Observer start...
2026-04-04 15:04:19,117 fail2ban.database [830]: INFO Connected to fail2ban persistent database '/var/lib/fail2
2026-04-04 15:04:19,119 fail2ban.jail [830]: INFO Creating new jail 'sshd'
2026-04-04 15:04:20,178 fail2ban.jail [830]: INFO Jail 'sshd' uses systemd {}
2026-04-04 15:04:20,209 fail2ban.jail [830]: INFO Initiated 'systemd' backend
2026-04-04 15:04:20,213 fail2ban.filter [830]: INFO maxLines: 1
2026-04-04 15:04:20,284 fail2ban.filtersystemd [830]: INFO [sshd] Added journal match for: '_SYSTEMD_UNIT=sshd.servi
2026-04-04 15:04:20,284 fail2ban.filter [830]: INFO maxRetry: 5
2026-04-04 15:04:20,284 fail2ban.filter [830]: INFO findtime: 600
2026-04-04 15:04:20,284 fail2ban.actions [830]: INFO banTime: 600
2026-04-04 15:04:20,284 fail2ban.filter [830]: INFO encoding: UTF-8
2026-04-04 15:04:20,296 fail2ban.jail [830]: INFO Jail 'sshd' started
2026-04-04 15:04:20,371 fail2ban.filtersystemd [830]: INFO [sshd] Jail is in operation now (process new journal entr
2026-04-04 15:56:45,514 fail2ban.server [830]: INFO Shutdown in progress...
2026-04-04 15:56:45,515 fail2ban.observer [830]: INFO Observer stop ... try to end queue 5 seconds
2026-04-04 15:56:45,536 fail2ban.observer [830]: INFO Observer stopped, 0 events remaining.
2026-04-04 15:56:45,577 fail2ban.server [830]: INFO Stopping all jails
2026-04-04 15:56:45,787 fail2ban.actions [830]: NOTICE [sshd] Flush ticket(s) with nftables
2026-04-04 15:56:45,788 fail2ban.jail [830]: INFO Jail 'sshd' stopped
2026-04-04 15:56:45,788 fail2ban.database [830]: INFO Connection to database closed.
2026-04-04 15:56:45,789 fail2ban.server [830]: INFO Exiting Fail2ban
2026-04-04 15:56:46,096 fail2ban.server [1535]: INFO -----
2026-04-04 15:56:46,096 fail2ban.server [1535]: INFO Starting Fail2ban v1.0.2
2026-04-04 15:56:46,098 fail2ban.observer [1535]: INFO Observer start...
2026-04-04 15:56:46,111 fail2ban.database [1535]: INFO Connected to fail2ban persistent database '/var/lib/fail
2026-04-04 15:56:46,112 fail2ban.jail [1535]: INFO Creating new jail 'sshd'
2026-04-04 15:56:46,317 fail2ban.jail [1535]: INFO Jail 'sshd' uses systemd {}
2026-04-04 15:56:46,318 fail2ban.jail [1535]: INFO Initiated 'systemd' backend
2026-04-04 15:56:46,319 fail2ban.filter [1535]: INFO maxLines: 1
2026-04-04 15:56:46,343 fail2ban.filtersystemd [1535]: INFO [sshd] Added journal match for: '_SYSTEMD_UNIT=sshd.servi
2026-04-04 15:56:46,343 fail2ban.filter [1535]: INFO maxRetry: 3
2026-04-04 15:56:46,343 fail2ban.filter [1535]: INFO findtime: 600
2026-04-04 15:56:46,344 fail2ban.actions [1535]: INFO banTime: 600
2026-04-04 15:56:46,344 fail2ban.filter [1535]: INFO encoding: UTF-8
2026-04-04 15:56:46,347 fail2ban.jail [1535]: INFO Jail 'sshd' started

```

Test de sécurité

```

donsam@donsam:~$ nmap 192.168.2.141
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-04 16:05 UTC
Nmap scan report for donsam (192.168.2.141)
Host is up (0.00034s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT      STATE SERVICE
2222/tcp  open  EtherNetIP-1

Nmap done: 1 IP address (1 host up) scanned in 0.20 seconds
donsam@donsam:~$

```

- Installation de Lynis

```
map: done: 1 of 6 addresses (1 host up): scanned in 1.157 seconds
donsam@donsam:~$ sudo apt update
Atteint :1 http://archive.ubuntu.com/ubuntu noble InRelease
Atteint :2 http://security.ubuntu.com/ubuntu noble-security InRelease
Réception de :3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Atteint :4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Réception de :5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1 875 kB]
Réception de :6 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1 663 kB]
3 664 ko réceptionnés en 45s (81,0 ko/s)
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Tous les paquets sont à jour.
donsam@donsam:~$ sudo apt install lynis -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les paquets supplémentaires suivants seront installés :
  menu
Paquets suggérés :
  apt-listbugs debsecan debsums tripwire samhain aide menu-l10n gksu | kde-cli-tools | ktsuss
Les NOUVEAUX paquets suivants seront installés :
  lynis menu
0 mis à jour, 2 nouvellement installés, 0 à enlever et 0 non mis à jour.
Il est nécessaire de prendre 602 ko dans les archives.
Après cette opération, 3 202 ko d'espace disque supplémentaires seront utilisés.
Réception de :1 http://archive.ubuntu.com/ubuntu noble/universe amd64 lynis all 3.0.9-1 [226 kB]
No VM guests are running detected hypervisor (qemu) binaries on this host.
donsam@donsam:~$ lynis --version
3.0.9
donsam@donsam:~$
```

- Audit complet

```
3.0.9
donsam@donsam:~$ sudo lynis audit system

[ Lynis 3.0.9 ]

#####
Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
welcome to redistribute it under the terms of the GNU General Public License.
See the LICENSE file for details about using this software.

2007-2021, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)
#####

[+] Initializing program
-----
- Detecting OS... [ DONE ]
- Checking profiles... [ DONE ]
- Detecting language and localization [ fr ]

-----
Program version: 3.0.9
Operating system: Linux
Operating system name: Ubuntu
Operating system version: 24.04
Kernel version: 6.8.0
Hardware platform: x86_64
Hostname: donsam
-----
Profiles: /etc/lynis/default.prf
Log file: /var/log/lynis.log
Report file: /var/log/lynis-report.dat
Report version: 1.0
Plugin directory: /etc/lynis/plugins
-----
Auditor: [Not Specified]
Language: fr
Test category: all
Test group: all
-----
- Program update status... [ PAS DE MISE A JOUR ]

[+] Outils système
-----
- Scanning available tools...
- Checking system binaries...
```

Analyse du rapport

```
donsam@donsam:~$ grep suggestion /var/log/lynis-report.dat
grep: /var/log/lynis-report.dat: Permission denied
donsam@donsam:~$ sudo grep suggestion /var/log/lynis-report.dat
suggestion[]=LYNIS|This release is more than 4 months old. Check the website or GitHub to see if there is an update available.|-|-|
suggestion[]=DEB-0280|Install libpam-tmpdir to set $TMP and $TMPDIR for PAM sessions|-|-|
suggestion[]=DEB-0810|Install apt-listbugs to display a list of critical bugs prior to each APT installation.|-|-|
suggestion[]=DEB-0811|Install apt-listchanges to display any significant changes prior to any upgrade via APT.|-|-|
suggestion[]=BOOT-5122|Set a password on GRUB boot loader to prevent altering boot configuration (e.g. boot in single user mode).|-|-|
suggestion[]=BOOT-5264|Consider hardening system services|Run '/usr/bin/systemd-analyze security SERVICE' for each service.|-|-|
suggestion[]=KRNL-5820|If not required, consider explicit disabling of core dump in /etc/security/limits.conf file|-|-|
suggestion[]=AUTH-9229|Check PAM configuration, add rounds if applicable and expire passwords to encrypt with new values.|-|-|
suggestion[]=AUTH-9230|Configure password hashing rounds in /etc/login.defs|-|-|
suggestion[]=AUTH-9262|Install a PAM module for password strength testing like pam_cracklib or pam_passwdqc|-|-|
suggestion[]=AUTH-9282|When possible set expire dates for all password protected accounts|-|-|
suggestion[]=AUTH-9286|Configure minimum password age in /etc/login.defs|-|-|
suggestion[]=AUTH-9286|Configure maximum password age in /etc/login.defs|-|-|
suggestion[]=AUTH-9328|Default umask in /etc/login.defs could be more strict like 027|-|-|
suggestion[]=FILE-6310|To decrease the impact of a full /home file system, place /home on a separate partition|-|-|
suggestion[]=FILE-6310|To decrease the impact of a full /tmp file system, place /tmp on a separate partition|-|-|
suggestion[]=FILE-6310|To decrease the impact of a full /var file system, place /var on a separate partition|-|-|
suggestion[]=USB-1000|Disable drivers like USB storage when not used, to prevent unauthorized storage or data theft|-|-|
suggestion[]=NAME-4028|Check DNS configuration for the dns domain name|-|-|
suggestion[]=PKGS-7370|Install debsums utility for the verification of packages with known good database.|-|-|
suggestion[]=PKGS-7394|Install package apt-show-versions for patch management purposes|-|-|
suggestion[]=NETW-3200|Determine if protocol 'dccp' is really needed on this system|-|-|
suggestion[]=NETW-3200|Determine if protocol 'sctp' is really needed on this system|-|-|
suggestion[]=NETW-3200|Determine if protocol 'rds' is really needed on this system|-|-|
suggestion[]=NETW-3200|Determine if protocol 'tipc' is really needed on this system|-|-|
suggestion[]=FIRE-4513|Check iptables rules to see which rules are currently not used|-|-|
suggestion[]=SSH-7408|Consider hardening SSH configuration|AllowTcpForwarding (set YES to NO)|-|-|
suggestion[]=SSH-7408|Consider hardening SSH configuration|ClientAliveCountMax (set 3 to 2)|-|-|
suggestion[]=SSH-7408|Consider hardening SSH configuration|LogLevel (set INFO to VERBOSE)|-|-|
suggestion[]=SSH-7408|Consider hardening SSH configuration|MaxSessions (set 10 to 2)|-|-|
suggestion[]=SSH-7408|Consider hardening SSH configuration|TCPKeepAlive (set YES to NO)|-|-|
suggestion[]=SSH-7408|Consider hardening SSH configuration|X11Forwarding (set YES to NO)|-|-|
suggestion[]=SSH-7408|Consider hardening SSH configuration|AllowAgentForwarding (set YES to NO)|-|-|
suggestion[]=LOGG-2154|Enable logging to an external logging host for archiving purposes and additional protection|-|-|
suggestion[]=BANN-7126|Add a legal banner to /etc/issue, to warn unauthorized users|-|-|
suggestion[]=BANN-7130|Add legal banner to /etc/issue.net, to warn unauthorized users|-|-|
suggestion[]=ACCT-9622|Enable process accounting|-|-|
suggestion[]=ACCT-9626|Enable sysstat to collect accounting (disabled)|-|-|
suggestion[]=ACCT-9628|Enable auditd to collect audit information|-|-|
suggestion[]=FINT-4350|Install a file integrity tool to monitor changes to critical and sensitive files|-|-|
suggestion[]=TOOL-5002|Determine if automation tools are present for system management|-|-|
suggestion[]=FILE-7524|Consider restricting file permissions|See screen output or log file|text:Use chmod to change file permissions.|-|-|
suggestion[]=KRNL-6000|One or more sysctl values differ from the scan profile and could be tweaked|Change sysctl values using sysctl-key>)|-|-|
suggestion[]=HRDN-7230|Harden the system by installing at least one malware scanner, to perform periodic file system scans.|-|-|
ntkit, OSSEC|
```

Amélioration de la sécurité

- Installation de libpam-tmpdir

```
ntkit, OSSEC|
donsam@donsam:~$ sudo apt install libpam-tmpdir
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les NOUVEAUX paquets suivants seront installés :
  libpam-tmpdir
0 mis à jour, 1 nouvellement installés, 0 à enlever et 0 non mis à jour.
Il est nécessaire de prendre 11,3 ko dans les archives.
Après cette opération, 52,2 ko d'espace disque supplémentaires seront utilisés.
Réception de :1 http://archive.ubuntu.com/ubuntu/noble/universe amd64 libpam-tmpdir amd64 0.09build1 [11,3 kB]
11,3 ko réceptionnés en 1s (14,3 ko/s)
Sélection du paquet libpam-tmpdir précédemment désélectionné.
(Lecture de la base de données... 127541 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de .../libpam-tmpdir_0.09build1_amd64.deb ...
Dépaquetage de libpam-tmpdir (0.09build1) ...
Paramétrage de libpam-tmpdir (0.09build1) ...
Traitement des actions différées (« triggers ») pour man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...
```

- Installation de apt-listbugs

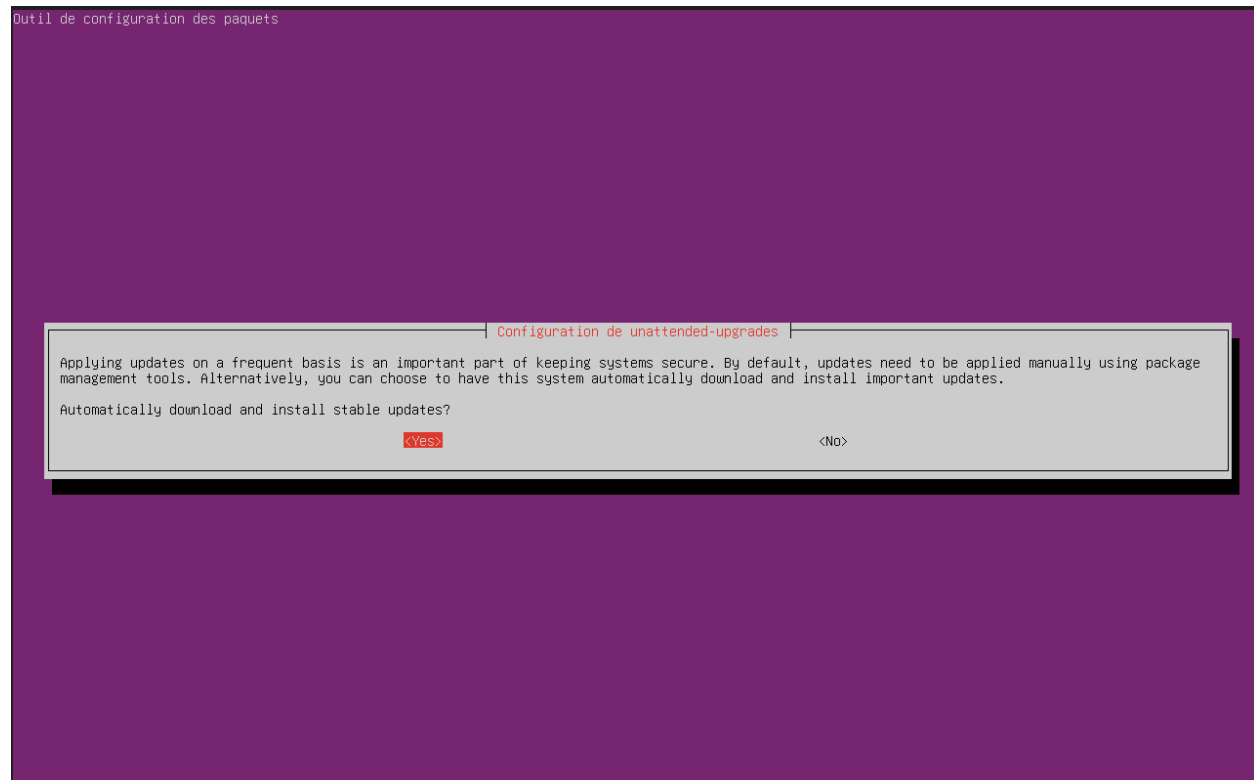
```
donsam@donsam:~$ sudo apt install apt-listbugs
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
```

- Installation de apt-listchanges

```
donsam@donsam:~$ sudo apt install apt-listchanges
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Paquets suggérés :
  default-mta | mail-transport-agent www-browser x-terminal-emulator
Les NOUVEAUX paquets suivants seront installés :
  apt-listchanges
0 mis à jour, 1 nouvellement installés, 0 à enlever et 0 non mis à jour.
Il est nécessaire de prendre 86,1 ko dans les archives.
Après cette opération, 397 ko d'espace disque supplémentaires seront utilisés.
Réception de :1 http://archive.ubuntu.com/ubuntu noble/main amd64 apt-listchanges all 3.27 [86,1 kB]
86,1 ko réceptionnés en 4s (22,2 ko/s)
```

- Installer le service

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
donsam@donsam:~$ sudo apt install unattended-upgrades -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
unattended-upgrades est déjà la version la plus récente (2.9.1+nmu4ubuntu1).
unattended-upgrades passé en « installé manuellement ».
0 mis à jour, 0 nouvellement installés, 0 à enlever et 0 non mis à jour.
donsam@donsam:~$
```



```
donsam@donsam:~$ sudo systemctl status unattended-upgrades
• unattended-upgrades.service - Unattended Upgrades Shutdown
  Loaded: loaded (/usr/lib/systemd/system/unattended-upgrades.service; enabled; preset: enabled)
  Active: active (running) since Sat 2026-04-04 15:04:17 UTC; 1h 28min ago
  Docs: man:unattended-upgrade(8)
  Main PID: 832 (unattended-upgr)
  Tasks: 2 (limit: 4601)
  Memory: 18.5M (peak: 18.9M)
  CPU: 510ms
  CGroup: /system.slice/unattended-upgrades.service
          └─832 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal

avril 04 15:04:17 donsam systemd[1]: Started unattended-upgrades.service - Unattended Upgrades Shutdown.
```

Installer un système de détection d'intégrité

```

donsam@donsam:~$ sudo apt install aide -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les paquets supplémentaires suivants seront installés :
  aide-common bsd-mailx libblockfile-bin libblockfile1 libmhash2 libnsl2 postfix ssl-cert
Paquets suggérés :
  figlet postfix-cdb postfix-doc postfix-ldap postfix-lmdb postfix-mta-sts-resolver postfix-mysql postfix-pcre postfix-sasl
  | dovecot-common
Les NOUVEAUX paquets suivants seront installés :
  aide aide-common bsd-mailx libblockfile-bin libblockfile1 libmhash2 libnsl2 postfix ssl-cert
0 mis à jour, 9 nouvellement installés, 0 à enlever et 0 non mis à jour.
Il est nécessaire de prendre 1 694 ko dans les archives.
Après cette opération, 5 574 ko d'espace disque supplémentaires seront utilisés.
Réception de :1 http://archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17,8 kB]
Réception de :2 http://archive.ubuntu.com/ubuntu noble/main amd64 libnsl2 amd64 2.0.0-2build1 [11,4 kB]
donsam@donsam:~$ sudo aideinit
Overwrite existing /var/lib/aide/aide.db.new [Yn]? y
Running aide --init...
[ 6725.469682] watchdog: BUG: soft lockup - CPU#1 stuck for 49s! [aide:63809]
Overwrite /var/lib/aide/aide.db [Yn]? y
donsam@donsam:~$ _

```

- Vérifier les permissions du dossier SSH

```

ERROR: Cannot open config file /etc/aide.conf : No such file or directory
donsam@donsam:~$ ls -ld ~/.ssh
drwx----- 2 donsam donsam 4096 mars  29 20:28 /home/donsam/.ssh
donsam@donsam:~$ chmod 700 ~/.ssh
donsam@donsam:~$ chmod 600 ~/.ssh/authorized_keys
donsam@donsam:~$ _

```

- Installation de logrotate

```

donsam@donsam:~$ sudo apt install logrotate
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
logrotate est déjà la version la plus récente (3.21.0-2build1).
logrotate passé en « installé manuellement ».
0 mis à jour, 0 nouvellement installés, 0 à enlever et 0 non mis à jour.
donsam@donsam:~$ _

```

Lancement de l'audit de sécurité

```
2007-2021, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)
#####

+) Initializing program
-----
- Detecting OS... [ DONE ]
- Checking profiles... [ DONE ]
- Detecting language and localization [ fr ]

-----
Program version: 3.0.9
Operating system: Linux
Operating system name: Ubuntu
Operating system version: 24.04
Kernel version: 6.8.0
Hardware platform: x86_64
Hostname: donsam
-----
Profiles: /etc/lynis/default.prf
Log file: /var/log/lynis.log
Report file: /var/log/lynis-report.dat
Report version: 1.0
Plugin directory: /etc/lynis/plugins
-----
Auditor: [Not Specified]
Language: fr
Test category: all
Test group: all
-----
- Program update status... [ PAS DE MISE A JOUR ]

+) Outils système
-----
- Scanning available tools...
- Checking system binaries...

+) Plugins (phase 1)
-----
Note : Les plugins ont des tests plus poussés qui peuvent prendre plusieurs minutes

- Plugin: debian
[
+) Debian Tests
-----
- Checking for system binaries that are required by Debian Tests...
- Checking /bin... [ FOUND ]
```

```

* Harden the system by installing at least one malware scanner, to perform periodic file system scans [HRDN-7230]
- Solution : Install a tool like rkhunter, chkrootkit, OSSEC
https://cisofy.com/lynis/controls/HRDN-7230/

Follow-up:
-----
- Show details of a test (lynis show details TEST-ID)
- Check the logfile for all details (less /var/log/lynis.log)
- Read security controls texts (https://cisofy.com)
- Use --upload to upload data to central system (Lynis Enterprise users)

=====

Lynis security scan details:

Hardening index : 69 [#####          ]
Tests performed : 262
Plugins enabled : 1

Components:
- Firewall           [V]
- Malware scanner    [X]

Scan mode:
Normal [V]  Forensics [ ]  Integration [ ]  Pentest [ ]

Lynis modules:
- Compliance status  [?]
- Security audit     [V]
- Vulnerability scan [V]

Files:
- Test and debug information : /var/log/lynis.log
- Report data                : /var/log/lynis-report.dat

=====

Lynis 3.0.9

Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)

2007-2021, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)

=====

[TIP]: Enhance Lynis audits by adding your settings to custom.prfl (see /etc/lynis/default.prfl for all settings)

```

? Questions

1. Pourquoi les mises à jour automatiques sont-elles importantes en production ?

En production, un système non patché est une cible facile. Les attaquants scrutent en permanence les serveurs exposés pour exploiter des vulnérabilités connues. Les mises à jour automatiques via `unattended-upgrades` permettent d'appliquer les correctifs de sécurité sans délai, sans attendre une intervention humaine. Elles aident aussi à maintenir la conformité avec des standards comme PCI-DSS ou ISO 27001. Il faut toutefois les encadrer : un patch mal testé peut casser un service critique en production.

2. Quel est le rôle d'un IDS comme AIDE ?

AIDE (Advanced Intrusion Detection Environment) surveille l'intégrité du système de fichiers. Concrètement, il calcule des empreintes cryptographiques (MD5, SHA256, SHA512...) de tous les fichiers système à un moment de référence, puis compare périodiquement l'état réel du système à cette baseline. Si un fichier est modifié, ajouté ou supprimé de manière non prévue, AIDE le signale. Ce n'est pas un outil de blocage, c'est un outil de **détection après coup**.

3. Pourquoi sécuriser les permissions des fichiers système ?

Des permissions trop ouvertes constituent l'une des premières portes d'entrée pour une escalade de privilèges. Par exemple, si `/etc/shadow` est lisible par un utilisateur ordinaire, ses mots de passe hashés peuvent être attaqués hors ligne. De même, un dossier `~/.ssh` avec de mauvaises permissions permet l'injection de clés non autorisées. Appliquer le principe du moindre privilège via `chmod`, `chown` réduit drastiquement la surface d'attaque interne.

4. Peut-on considérer un serveur totalement sécurisé ?

Non, la sécurité absolue n'existe pas. Chaque composant logiciel ajouté (dépendance, protocole, service) introduit une nouvelle surface d'attaque potentielle. Les vulnérabilités zero-day sont par définition inconnues jusqu'à leur divulgation publique. Et même un système techniquement bien configuré reste vulnérable au facteur humain : une mauvaise manipulation, une clé SSH mal gérée, ou une attaque par ingénierie sociale peuvent tout compromettre. L'objectif réaliste est de **minimiser les risques** et de **détecter rapidement** toute anomalie, comme le montre l'amélioration du Hardening Index de 67 à 68 après les corrections appliquées dans ce TD.

Conclusion

À travers ce TD, j'ai pu mettre en pratique de manière concrète les notions de sécurisation d'un serveur Linux. Configurer SSH, installer Fail2Ban, paramétrer UFW et interpréter un rapport Lynis m'a permis de comprendre réellement comment fonctionne la défense en profondeur d'un système. Voir le Hardening Index passer de 67 à 68 après mes corrections a été une satisfaction directe de mon travail. Je retiens surtout qu'un serveur ne sera jamais totalement sécurisé, mais que chaque amélioration compte et que la vigilance doit être permanente.